
SI 339 Winter 2025 Final Project

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Date: April 20, 2025

Project Purpose

We want you to spend **18 - 25 hours** working on an independent project*. The topic is your choice but it either needs to be a coding project or an accessibility project.

Resources:

- [Final Project Overview](#)
- [Final Project Grading Rubric](#)
- [Photoshop cheatsheet for resizing images for web](#)

**You can work with a partner, but we would expect that your final project would take 36-50 hours to complete and that you would each turn in your own report with your hours logged, progress, and lessons learned. If you are working with a partner [please fill out this form](#).*

Contact:

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URL's

- **Final Project Presentation Slides Link:** https://docs.google.com/presentation/d/1h_3WeEDcCzZWZCoRdjh0J6X-GIXuUwXOMg72rpCTa0s/edit?usp=sharing
- **Published Website URL:** <https://scodegithub1234.github.io/milk-and-froth/>
- **Github Repo URL:** <https://github.com/scodegithub1234/milk-and-froth>

Our Objectives:

Explain your project in 2-3 sentences below:

My project is a website revamp for Milk and Froth, a small ice cream business in Ann Arbor in need of a more modern, accessible online presence. The goal was to enhance both the visual appeal and the usability of the website by improving color contrast, updating font for readability, fixing dead links, and introducing new features that support a better user experience. This included focusing on accessibility best practices to ensure the site is more inclusive for all users, especially those with visual impairments.

Project Type (Pick Coding or Accessibility):

Explain the scope of your project - how did you expand upon what you learned in class this semester. What was original/unique about this project compared to what we did in class?

This project allowed me to take the foundational knowledge from class—such as basic HTML, CSS, and accessibility principles—and apply it to a real-world situation where I had to evaluate and improve an existing website. While class projects were more structured with clear guidelines, this project required me to think more independently and critically, especially when identifying areas for improvement and implementing design changes that were not specified to be changed anywhere. What made this project unique was the opportunity to actually collaborate with a real company's aesthetic and identity, while also ensuring accessibility for all types of users that may come across the website. Another unique aspect was that in class we only did accessibility audits for local businesses. To be able to look at a website online and use an accessibility audit to improve the website is something we talked about doing in class but never implemented. In this project I was able to see the full process through and improve on an existing website and incorporate my own design changes.

Where did you feel like you spent the majority of your time when working on this project?

The majority of my time was spent on the redesign process, especially when it came to testing and adjusting design elements for accessibility. I worked on refining the color palette to ensure proper contrast ratios for text and background elements, which involved a lot of trial and error. I also devoted significant time to cleaning up and updating the website structure—removing dead links, optimizing navigation menus, and ensuring that all headings and labels were clearly defined for assistive technologies.

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Making the website keyboard-navigable and ensuring that interactive elements like buttons and forms had proper ARIA labels also took a great deal of focus and testing. I also had to make sure that I was coding from scratch before looking at the website for reference to its material to ensure that my website was completely original in design and only followed the information format and contained Milk & Froth's details without copying their design, so setting up the website before referencing the existing website took a lot of creative consideration.

What were areas that you found challenging for your project? How did you navigate these challenges? What were some of your lessons learned completing the project?

One of the most challenging aspects of the project was finding the right balance between maintaining the brand's aesthetic and implementing necessary accessibility standards. For example, the original brand colors were visually appealing but didn't meet WCAG contrast requirements, so I had to propose adjusted shades that still felt on-brand but were much easier to read. Navigating these changes required flexibility as well as research into accessibility. Another challenge was making sure the site worked well for users relying on screen readers or those with mobility impairments—things like properly labeling forms, using semantic HTML, and ensuring that navigation could be done entirely via keyboard.

From this experience, I learned how small design and coding decisions can have a significant impact on user experience, particularly for those with disabilities. I learned that accessibility is an ongoing design process and although I made many changes now in the future there will be more considerations to be made based on new accessibility guidelines and features. I learned the importance of testing with different tools and perspectives.

For coding projects only:

Explain how you ensured that your site is fully accessible and usable for all - including those with different disabilities.

To ensure the site was accessible and usable for individuals with various disabilities, I followed the Web Content Accessibility Guidelines and implemented several inclusive design practices. I used semantic HTML to ensure that screen readers could accurately interpret the structure and content of each page. I made sure all images included descriptive alt text, and that buttons, links, and form inputs had clear labels and accessible names.

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I also tested the site's color contrast using contrast checker tools to meet accessibility thresholds, making the text easier to read for users with visual impairments or color blindness. I edited the font to ensure that it would not be hard for people with impairments to read. For users with motor impairments, I ensured the entire site could be navigated using a keyboard alone, with focus indicators that are clearly visible when tabbing through elements. Additionally, I avoided using elements that relied solely on color to convey information and made sure all interactive features had enough time for users to respond. Throughout development, I used accessibility evaluation tools such as WAVE and aXe to identify issues and correct them. By continually testing the site from different accessibility perspectives, I was able to create a user-friendly website that accommodates a wide range of needs.

If you worked with a partner:

Explain what your individual contributions to the project were, what were you able to do together in your extended hours that you would not have been able to complete on your own.

N/A

Project Time:

- **Weekly Progress 1 (Dates)**
 - Hours spent: 4
 - Challenges:
 - Reviewing the original Milk and Froth website to identify accessibility issues was more time-consuming than expected. I referenced my accessibility audit from HW 4.
 - Learning the color palette of the website design
 - Successes:
 - Created a checklist of issues including low color contrast, missing alt text, and keyboard navigation problems.
 - Set up the project structure for the new version of the site.
- **Weekly Progress 2 (Dates)**
 - Hours spent: 6
 - Challenges:

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- Balancing visual design with accessibility standards, particularly in choosing new colors that looked good and matched the theme but also met contrast requirements.
- Replacing certain UI elements that weren't semantic from the original website such as unnecessary <div>s and <header>s.
- Successes:
 - Updated the color scheme to be accessible while still reflecting the brand identity.
 - Rewrote and structured the homepage HTML using semantic tags.
 - Added descriptive alt text for all images and ensured keyboard accessibility on pages.
- **Weekly Progress 3 (Dates)**
 - Hours spent: 3
 - Challenges:
 - Implemented and tested various JS functionalities such as the forms.
 - Ensuring consistency across different screen sizes and devices with mobile and laptop views.
 - Successes:
 - Verified keyboard navigation for the entire site and fixed several focus issues.
 - Made the new features (like an updated contact form) fully accessible.
- **Weekly Progress 4 (Dates)**
 - Hours spent: 7
 - Challenges:
 - Final round of testing uncovered accessibility bugs on mobile view and laptop view.
 - Making sure all links and navigation menus were consistent.
 - Successes:
 - Resolved all remaining bugs aXe and WAVE audits.
 - Finalized and deployed the updated website design.

Appendix

You might include here resources and references you used in completing your final project whether it's a coding or accessibility project.

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- Milk & Froth Website: <https://www.milkandfroth.com/>
- Color Contrast Checker: <https://webaim.org/resources/contrastchecker/>
- WAVE and aXe web development tools (in web browser)
- WCAG contrast checker: <https://accessibleweb.com/color-contrast-checker/>
- Went to Milk & Froth onsite Ann Arbor location to see current menu